ClaimIQ

Pre-submission claim audit for Indian hospitals

Catch the deduction before the insurer does.

A hospital submits a claim packet to a TPA. Weeks later money comes back — usually less than was billed — with a deduction sheet nobody on the billing desk fully understands. By then the patient has gone home, the bill is closed, and whatever was disallowed is a fight or a write-off. ClaimIQ reads the packet **before** it is submitted and splits the bill three ways, citing the rule behind every rupee it moves.

Engine	Python 3.12 · LangGraph · Pydantic v2 · FastAPI
Rule corpus	104 cited rules · unverified-97f8583129
Product UI	Static SPA, no build step, served by the API
Tests	unknown passing
Source	~12,997 lines of Python

Generated 02 August 2026 from a live run of this repository. Every figure in this document was measured at build time, not transcribed.

Estimates, not adjudications. All data in this build is synthetic.

1 — The problem

Indian health insurance does not deny a claim outright very often. It **shaves** it. The deduction sheet arrives weeks after discharge, itemised in a vocabulary the billing desk did not write, and by then nobody can do anything about it.

Two losses, and only one of them is visible

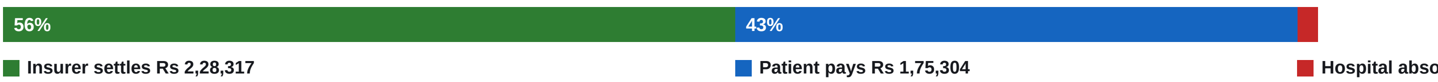
The patient learns their out-of-pocket number at the discharge counter instead of at admission. The largest single cause is the proportionate deduction: choose a room above the policy's room-rent cap and the insurer scales down not just the room charge but a whole basket of associated charges — surgeon's fee, OT, nursing, anaesthesia — by the same ratio.

The hospital loses money it never notices. The non-payable framework is not one list, it is four, and they differ by who bears the cost. List I items are the patient's. Lists II, III and IV are charges already deemed covered by the room rate, the procedure package or the cost of treatment — the hospital cannot bill them to the insurer *or* to the patient.

> A List I hit means *warn the patient*. A List II/III/IV hit means *your billing master is wrong and you are losing this on every single claim you file* — the same items, forever, across thousands of claims. This distinction is the product. A blocklist that says “these items are non-payable” surfaces only the first finding. The second is worth far more and is invisible without modelling who bears the cost.

What that costs on one real claim

The cardiac sample in this repository: a CABG admission, gross Rs 4,10,000, eight days in a room billed at Rs 12,000/day against a Rs 6,000/day policy cap. Audited offline, with no language model configured, so these figures are reproducible by anyone with the repository:



	Amount	What it is
Gross bill	Rs 4,10,000	What the hospital billed
Insurer settles	Rs 2,28,317	After caps, sub-limits and co-pay
Patient pays	Rs 1,75,304	Optional items plus policy deductions
Hospital absorbs	Rs 6,380	Already covered by room or procedure charges

The room-rent breach alone accounts for Rs 1,45,685 of that — and none of it was knowable to the family at admission, which is precisely when it could have been avoided by choosing a cheaper room.

2 — What the product does

Upload the packet you already assemble for the TPA. ClaimIQ reads it, asks only for what no document stated, and returns three numbers with the working shown.

Shaded steps are where a human decides. Everything between them is deterministic or cited.

OUTPUT — THE SAME RUPEE NEVER COUNTED TWICE

Insurer settles
after caps, sub-limits, co-pay

Patient pays
optional items + policy deductions

Hospital absorbs
already covered by room / procedure

The four screens

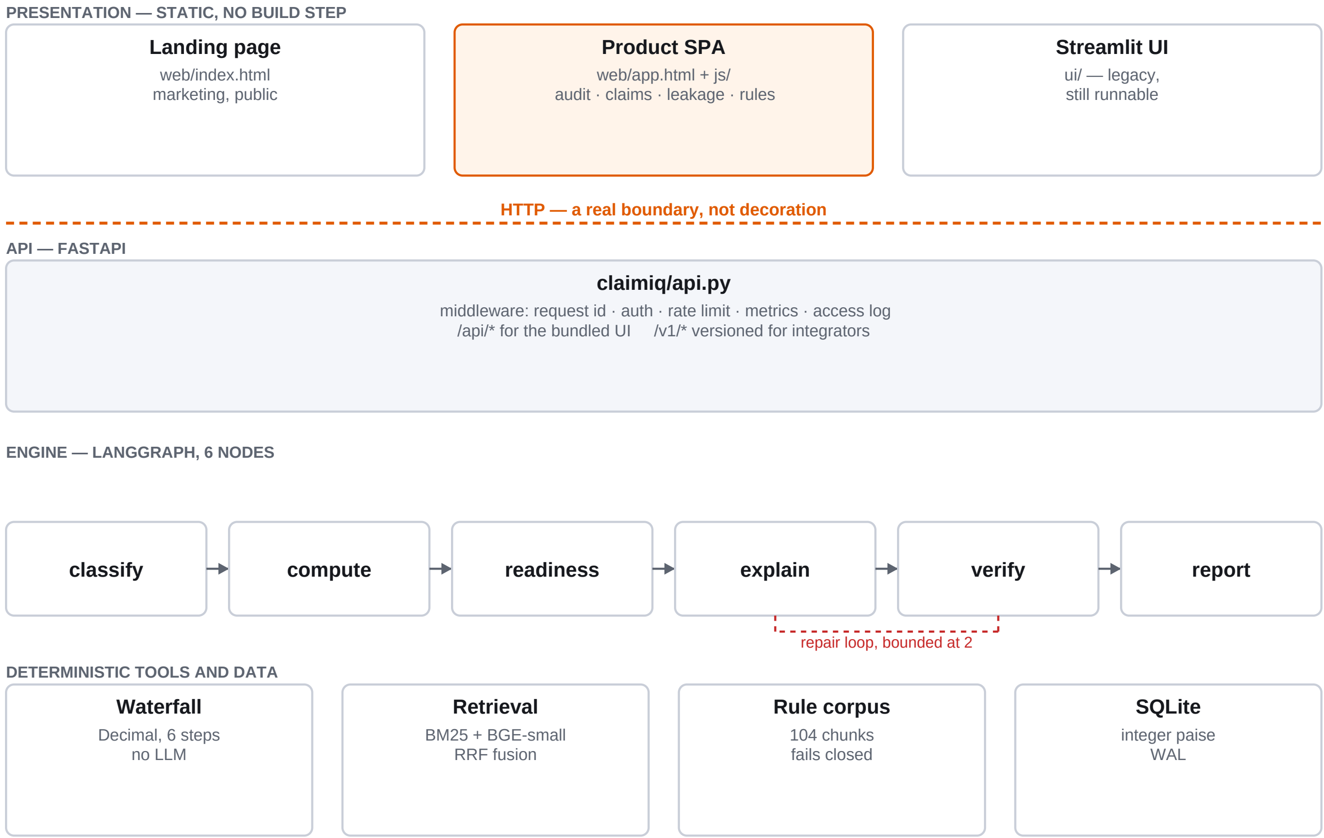
Screen	What it answers
Check a claim	What will this specific packet lose, and why — every finding expandable to the rule, the citation and the arithmetic.
Claims	What have we audited, and what did it say — searchable history, reopenable.
Leakage	Which billing mistakes cost the most across the whole book, ranked by aggregate loss.
Rule catalog	What does the tool actually believe, and on whose authority.

Three states, and the difference between them

A claim comes back as **Ready to submit**, **Needs attention**, or **Could not fully check this claim**. The third exists because “we found nothing” and “we could not look” are different answers, and conflating them is how a tool becomes dangerous. Coverage is measured in rupees, not lines: one unreadable Rs 90,000 line matters more than nine unreadable Rs 10 lines.

3 — Architecture

Four layers with one hard boundary. The UI never imports the engine; it speaks HTTP, which is what makes the Streamlit UI replaceable by a static SPA without touching a single node.



Why the LLM does no arithmetic

This is the load-bearing decision. The agent calls a deterministic `Decimal` calculator for every rupee, and the model only describes numbers it was handed. That is not a limitation worked around — it is what makes the verifier possible.

> The model has judgment; the tool has precision.

An invariant is asserted on every single audit, and a narrative containing a figure the calculator never produced is rejected and regenerated:

```
projected_settlement + patient_liability + hospital_writeoff == gross_bill
```

The six-node graph

Node	Responsibility
classify	Each line item → payable, one of four non-payable lists, or unmapped. Must cite the rule it decided from.
compute	Deterministic waterfall across three insurer profiles.
readiness	Twelve conditional document requirements, nine consistency checks.
explain	Narrative and prioritised actions, from a fixed fact block.
verify	Three guardrails: the arithmetic invariant, citation coverage, and narrative-to-figure reconciliation.
report	Routing target for the verifier's conditional edge.

The backward edge from verify to explain is why this is a graph and not a chain. It is bounded at two repairs, so a model that cannot be talked into consistency degrades visibly instead of looping.

4 — Where the rules come from

A rule that cannot name its authority is not a rule, it is an opinion. The loader fails closed: a chunk missing a citation, a severity or a resolvable source stops the process rather than becoming a determination on somebody's claim.

List	Rules	Who bears it	Source
List I — optional items	35	Patient	IRDAI/HLT/REG/CIR/176/09/2019, Annexure I
List II — subsumed into room	24	Hospital	IRDA/HLT/REG/CIR/146/07/2016, Annexure II
List III — subsumed into procedure	19	Hospital	IRDA/HLT/REG/CIR/146/07/2016, Annexure III
List IV — subsumed into treatment	16	Hospital	IRDA/HLT/REG/CIR/146/07/2016, Annexure IV
Policy wording	10	—	Synthetic, labelled as such

Honesty constraints, enforced by tests

• **Citation precision.** The four lists correspond to Annexures I–IV as a whole; that mapping is verifiable and is asserted. No individual item claims an annexure line number, because no item has been matched against one. A test fails if any rule claims item-level precision.

• **Verification status.** Every source ships unverified, and the corpus version string degrades its own prefix to `unverified-` while that is true. The stamp cannot claim more confidence than its weakest source.

• **Currency is unresolved.** Whether the 2024 IRDAI Master Circular supersedes the 2016/2019 guidelines has not been confirmed. The product does not assert it either way, and shows a standing notice on every audit.

> Resolving that question is the single highest-value item outstanding. It decides whether the corpus rests on live regulation or on a historical snapshot.

5 — What has been measured

Against 87 hand-labelled line items written to imitate real hospital bill printing. Read accuracy against the `head_only` control, which ignores the description entirely — the labelled set's billing head nearly determines the payable split by itself, so overall accuracy is inflated for every strategy.

Strategy	Accuracy	Non-payable recall	False deduction	Cited
head_only	51.7%	30.5%	3.6%	0%
keyword	82.8%	74.6%	0.0%	100%
retrieval	40.2%	11.9%	0.0%	100%
deterministic	81.6%	74.6%	0.0%	100%

Non-payable recall is the honest number. It measures picking the correct list among four, which the billing head cannot indicate at all.

False deduction rate is the error that matters most. Missing a non-payable item costs the hospital a recovery opportunity; wrongly disallowing a real medical charge produces a bill the patient should never have been shown. The retrieval gate sits where that rate is zero, and the lost recall is what the language model is there to recover.

What the numbers do not say

Roughly one non-payable item in four is missed with no model configured. The product states this on every audit rather than presenting an unmatched line as a clean one. There are no deployment statistics in this document — no rupees recovered, no claims processed, no hospitals live — because there are none to report yet, and a plausible-looking estimate would be worse than a blank.

6 — Why this can win

It sells against a number nobody else shows

Every competing tool in this space stops at “these items are non-payable”. That is a compliance feature: it prevents a deduction the hospital was going to eat anyway. ClaimIQ models the *bearer*, which turns the same rule set into a recurring-revenue argument: *this line costs you money on every claim you file, here is the rule, here is the aggregate across your book*. The buyer is not buying compliance, they are buying a leak they did not know they had.

The output is checkable, which is what closes a hospital sale

Hospital procurement does not evaluate on demo polish, it evaluates on whether the finance office can defend a number to a TPA. Every finding carries the rule, the citation, the field that failed, what was found, what was expected, and the arithmetic. A billing manager can copy one finding into an email and argue it. A tool whose output cannot be argued with is a tool that gets used once.

The honesty is a moat, not a liability

The product says out loud that its corpus is unverified, that the offline path misses a quarter of non-payable items, that scan text reaches a third-party model, and that there is no multi-tenancy. Competitors claiming 99% accuracy cannot survive the first question from a compliance officer. Being the vendor who already answered it is a durable position — and every one of those statements is a roadmap item with a known fix.

Timing

The IRDAI Master Circular of May 2024 consolidated 55 circulars and pushed standardisation and cashless settlement hard. Deduction disputes are being formalised at exactly the moment hospitals have the least tolerance for unexplained write-offs.

Honest competitive risk

The rule corpus is not defensible on its own — the annexures are public. The defensibility is in the bearer model, the citation discipline, the extraction path that reads a real hospital bill, and the portfolio view that turns single claims into a billing-master fix. A TPA or a large HIS vendor could build this. The bet is that they will not build it *honestly*, because their incentive is to sell certainty.

7 — What is built, and what is not

Stated plainly, because the gaps decide whether this can be deployed and the strengths decide whether it is worth deploying.

Built and tested

Area	State
Deduction engine	Six-step waterfall, three insurer profiles, exact Decimal, three-bucket invariant asserted on every audit.
Rule source	104 cited rules, fail-closed loader, source registry with published/retrieved/verified dates.
Verifier	Arithmetic invariant, citation coverage on every deduction, narrative reconciliation, bounded repair loop.
Extraction	Native PDF table parse (exact, no model), on-device OCR + model for scans, policy schedule and discharge summary reading.
Product UI	Static SPA: audit, claims history, leakage dashboard, rule catalog. Shareable URLs, print stylesheet, keyboard navigable.
Persistence	SQLite, integer paise, WAL, searchable claim history.
Tests	unknown passing, offline and deterministic.

Not built — and why it matters

Gap	Consequence
Rule currency unverified	Sources not checked against the issuing body's current publication (Guidelines on Standardization in Health Insurance; Modif
Item-level citations	Citations are list-level. No item has been matched line-by-line against the official annexure.
PHI to third-party model	Scan text — which carries clinical detail — is sent to the configured provider. Native PDFs never leave the machine.
Vision extraction	Present but not viable on a free tier; a page costs more tokens than the per-minute allowance.
Real-world validation	Measured only against synthetic, hand-labelled data. No accuracy claim against real adjudicated claims exists.

The order I would fix them in

1. Confirm whether the 2024 Master Circular supersedes the 2016/2019 guidelines. Nothing else matters if the rules are stale.
2. Verify the four annexures item by item and upgrade citation precision.
3. Validate against a real book of adjudicated claims — the only number a hospital will actually believe.
4. Decide the PHI question in writing before any pilot touches real patients.

This document was generated from a live run of the repository. If a figure here disagrees with the product, the product is right and this file is stale — regenerate it with `python scripts/gen_project_doc.py`.